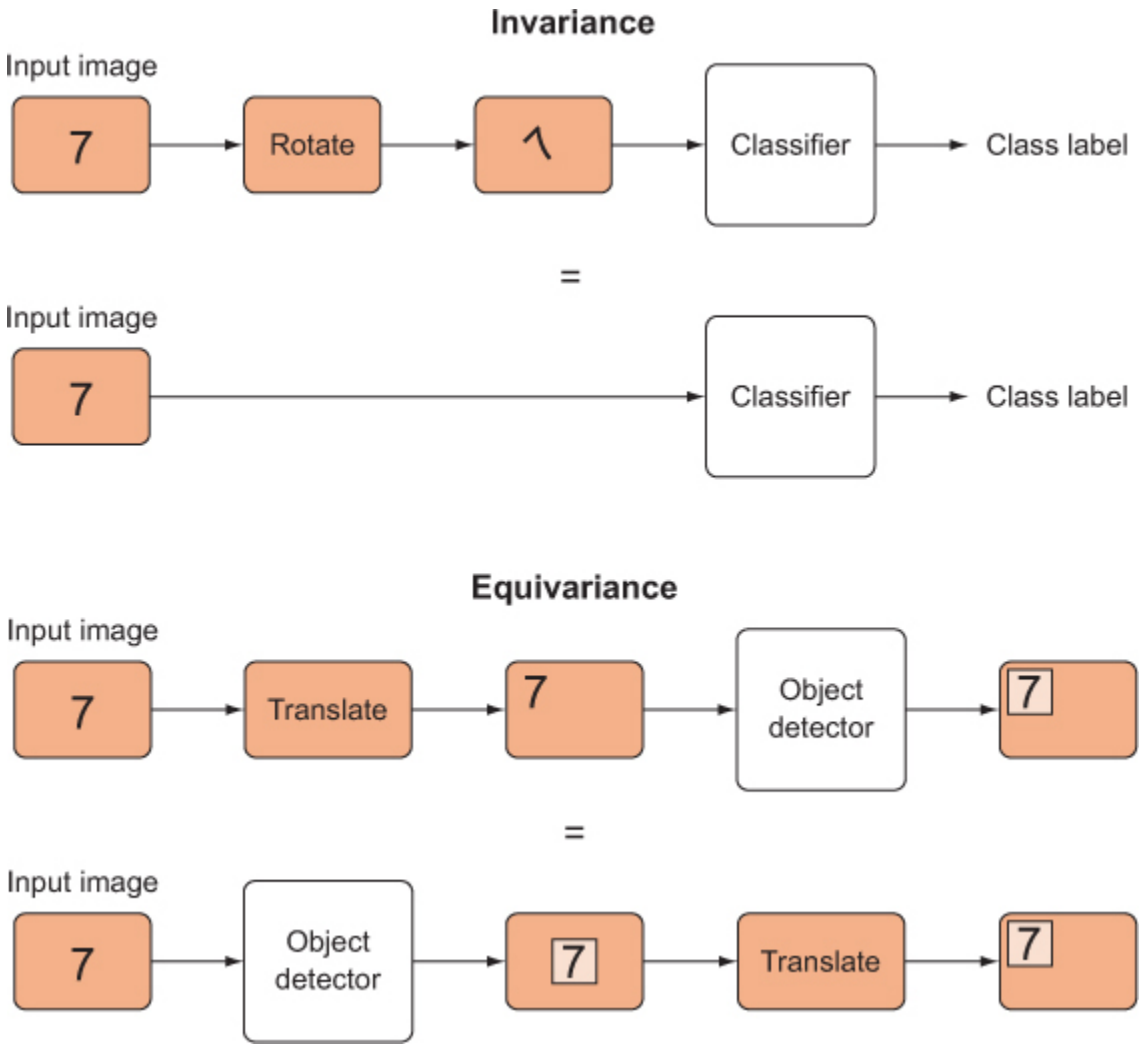


If we use a CNN to detect the location of an object in an image, it is no longer invariant to translation but rather *equivariant* (bottom panel of figure 10.3). *Equivariance* is when $f(g(x)) = g(f(x))$, for some transformation function g. This equation says that if we take an image with a face in the center, apply a translation so the face is now in the top-left corner, and then run it through a CNN face detector, the result is the same as if we had just run the original centered image through the face detector and then translated the output to the top-left corner. The distinction is subtle, and often invariance and equivariance are used interchangeably since they are related.

Ideally, we want our neural network architectures to be invariant to many kinds of transformations our input data might suffer. In the case of images, we generally want our machine learning model to be invariant to